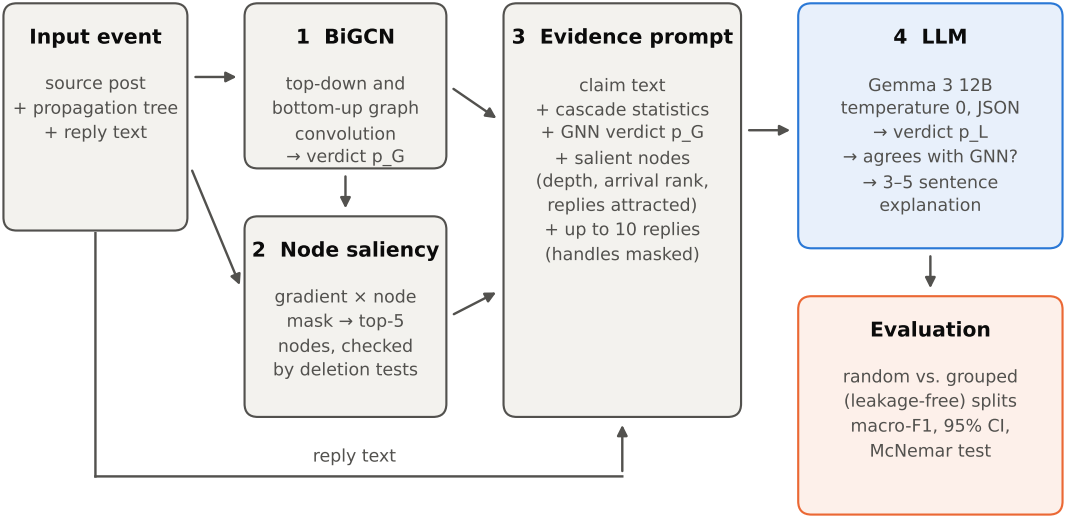
Input event
source post
+ propagation tree
+ reply text
1 BiGCN
top-down and
bottom-up graph
convolution
→ verdict p_G
2 Node saliency
gradient × node
mask → top-5
nodes, checked
by deletion tests
3 Evidence prompt
claim text
+ cascade statistics
+ GNN verdict p_G
+ salient nodes
(depth, arrival rank,
replies attracted)
+ up to 10 replies
(handles masked)
4 LLM
Gemma 3 12B
temperature 0, JSON
→ verdict p_L
→ agrees with GNN?
→ 3–5 sentence
explanation
Evaluation
random vs. grouped
(leakage-free) splits
macro-F1, 95% CI,
McNemar test
reply text